

Improving Design System Adoption with Inner Source

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Abstract

This study investigates the challenges in adopting SAP's design system, SAP Fiori, across various teams within the organization. Despite its maturity, adoption has been limited due to knowledge gaps, slow system evolution, lack of flexibility, and inefficiencies. A case study involving 76 stakeholder interviews identified these barriers and proposed solutions, including improved documentation, modular technical infrastructure, and an inner source model for shared ownership of the design system. Lessons learned emphasize the value of empowering teams to take ownership, which improved motivation and collaboration. However, capacity and skill gaps, particularly in maintaining high-quality components, remain a challenge. Scaling the design system expertise and refining key roles, like design system experts, will be critical for sustaining adoption and fostering a stronger design community.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Interactive systems and tools; User interface management systems; Interaction design; Interaction design process and methods; User interface design.

Keywords

Design System, Design Operations, User Experience, Inner Source

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1 Introduction

SAP is one of the largest global software vendors for enterprise applications. SAP solutions address all major segments of the global economy and with that offer a vast set of software products for all domains. As different parts of the portfolio have been developed at different points in time including acquisitions it is a continuous effort harmonizing and integrating the user experience across the portfolio. An important part of this endeavor is the global design system, named *SAP Fiori*¹, which has been evolving into one of

the most mature systems for enterprise application design over the past twelve years.

1.1 Problem overview

Despite the maturity of the design system and tools and processes that have been established to support product designers applying it, it could be observed in regular product reviews and customer feedback that the adoption could still be improved. The fact that the design system already was very comprehensive (more than 270 components for web-based applications, more than 80 for native mobile applications on iOS and Android) gave an indication that the potential of improvement would not be found within the design system itself but rather within the supporting infrastructure and processes.

1.2 Purpose of this study

The goal of the study is, to identify the main inhibitors to the full uptake of the design system. To better understand the reasons that prevented product design teams from a broader adoption of the design system a comprehensive study was planned. This study should include the perspectives of all stakeholders in the process and should direct future improvements of the design system and the supporting infrastructure and processes.

Acknowledging the various roles in the product design and development cycle, the objective is not only to understand the situation for the design teams but also for other stakeholders like engineering, product management, sales and others. As a result, a set of measures should be identified to improve the uptake of the design system in future. Issues associated with adoption have been reported by other software vendors from various domains as well [1], [2], so the intention of this research also is, to share insights, helping other design organizations to learn and build on this.

2 Background

2.1 Related work

Design systems provide single source of truth and assure consistency in cross-functional teams [3], [4], [5]. When talking about design systems we are not only referring to component libraries, but a collection of documented elements that embody an organization's design rules and principles [5]. Although this way of working is a prominent way to standardize the intersection of design and development within an enterprise [6], there is a noticeable gap in research concerning synthesizing knowledge on the subject [7] largely due to industry practices outpacing academic instruction [5]. To bridge this gap, and support body of knowledge within the field of HCI, we propose insights into infrastructure and processes surrounding practical applications of design system in large international software vendor, SAP SE.

¹<https://www.sap.com/products/technology-platform/fiori.html>

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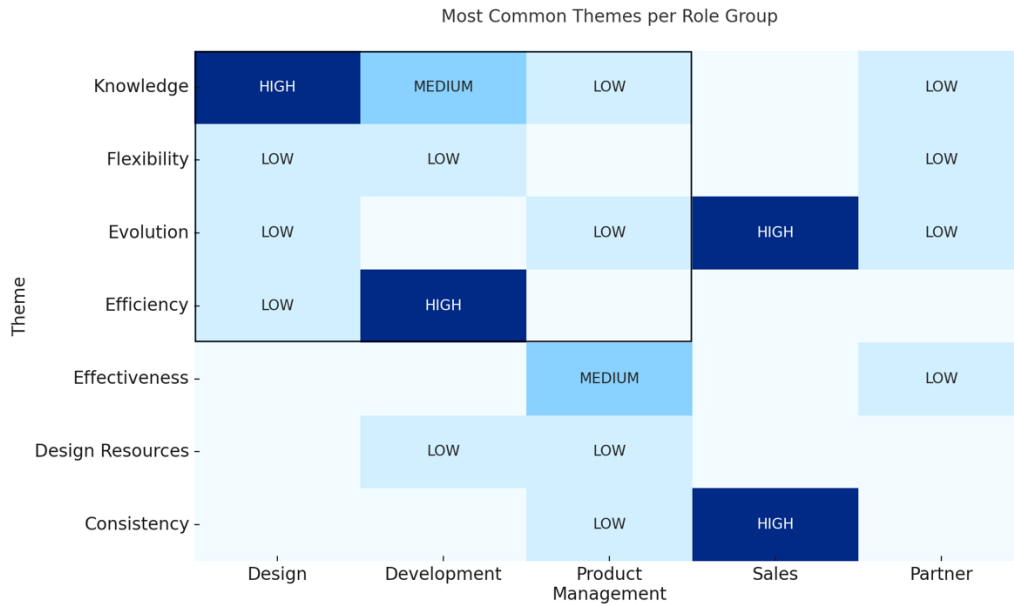


Figure 1: Heatmap illustrating the most common themes identified per role group. The color intensity represents the frequency of mentions, with "HIGH," "MEDIUM," "LOW," and empty reflecting the significance of each theme as reported by different roles, including Design, Development, Product Management, Sales, and Partner.

2.2 Design System at SAP

Building elements of the SAP design system begin with establishing foundational guidance for creating content, which includes detailed design specifications for components, patterns, floorplans, and assets like fonts and icons. This is followed by providing comprehensive enablement resources such as UI kits, technology frameworks, and marketing templates to support designers and developers in adopting and implementing the system effectively.

2.3 Challenges in adoption

As most large organizations, development is happening in different lines of business being responsible for a certain part of the portfolio. Central teams deliver infrastructural parts like platforms, services, but also the design system that are supposed to reduce development cost via scaling effects. The uptake of these central deliveries is moderated by other influences like timelines, economic, and customer requirements so that central deliveries always compete with local tailored solutions. To fully understand inhibiting factors to increase adoption of the design system and reduce the number of local solutions that eventually will become design debt is therefore of high importance.

3 Methodology

3.1 Case study

To explore the challenges of design system adoption across various stakeholders at SAP, we applied a case study approach. We employed Gerring's understanding of it as "an intensive study of a single unit with an aim to generalize across a larger set of units" [8]. We focused on SAP's design system adoption challenges across

multiple lines of business as our unit of analysis. The aim was to identify and organize the challenges, which could then be explored through expert interviews.

3.2 Stakeholder interviews

We interviewed 76 participants from design, development, product management, sales, and implementation partners. These stakeholders interacted with design systems in various ways, from product design to customer interactions. Sales served as a proxy for customer perspectives, summarizing key interaction issues. The semi-structured interviews used Mural² collaboration boards, allowing participant to add or refine the discussion points, ensuring a stakeholder-centered dialogue.

The interviews were not recorded; responses were documented in collaboration boards and meeting notes. We have used a thematic analysis, to identify recurring patterns and themes across our interviewee responses based of the Braun and Clarke's approach [9]. A total of 265 problem statements were derived, clustered by similar semantics, and categorized by stakeholder role. The most frequent issues per role were further analyzed and grouped into recurring themes, as shown in Figure 1.

3.3 Expert interviews

To validate the themes identified from the problem statements gathered in the first interview, we conducted a second set of six interviews with internal experts from our primary stakeholder groups. We chose all interview participants using an expert sampling strategy to make sure they had deep knowledge and hands-on

²<https://www.mural.co/>

experience with the topic [10]. These semi-structured interviews included both a structured part for theme validation and an open part for discussing new topics. Conducted via Microsoft Teams, each session lasted around 45 minutes. With interviewees consent, audio was recorded, transcribed, and notes were shared to ensure alignment on discussed topics.

While the initial stakeholder interviews offered a broader perspective, the later expert interviews helped refine our understanding of the problem and identify key areas for improvement. Experts confirmed the validity and priority of the themes but noted that some were general design issues rather than specific to design system adoption. Specifically, consistency was considered a result of suboptimal adoption rather than a direct inhibitor, and challenges like lack of design resources and prioritization were deemed organizational issues rather than related to the design system adoption itself. As a result, we continued with our focus on the remaining four main themes: knowledge, flexibility, evolution, and efficiency.

3.4 Ethical consideration

In this study, we followed ethical guidelines for research involving human participants (see A.1). Ethical approval was obtained from the SAP internal review board, and informed consent was obtained from all interviewees before participation. They were informed of their right to withdraw at any time, and all data was anonymized to ensure privacy.

4 Results

As a result of our analysis and the validation with the experts, we were able to identify the following four main themes that inhibit the adoption of the design system in our organization.

4.1 Knowledge

The most important inhibitor for all three internal stakeholder groups is knowledge. This includes knowledge of what elements and building blocks are available (inventory) and how they should be used to cater for certain use cases (guidelines) as the most common aspects mentioned in both sets. The availability of features in different technologies used by different products was another important aspect (mapping). In addition to that, stakeholders expressed the wish to better understand the reasoning behind certain decisions that were taken during the definition of the design system and how contributions to the design system could be made, “Design System is a shared element and needs a defined collaboration model. I don’t believe in the centralization. What matters is what’s in it for me and how I can contribute.” The main reasons for that knowledge gap are the complexity and volume of the documentation and the fact that different resources must be used to get a full picture. Integration of design and technical documentation was not sufficient. In many cases, documentation was not always up to date with the actual state of development. As one product developer observed, “Technology documentation lags behind implementation: sometimes desired features are actually available, but they are not documented yet.”

Looking at this from an organizational perspective, we can find that training and documentation are a strategic, long-term, cross-organizational, and often underestimated investment. Making this

the responsibility of a single team holds the risk of failure due to capacity, expertise, and visibility. Setting up and securing such investment is challenging as it requires continuous and reliable support from various parties across organizational boundaries and cost-centers.

4.2 Evolution

The second most important issue is the speed of evolution of the design system. Product teams requesting for new capabilities are always in competition for scarce resources. Especially in a set up where new design system features are delivered by central teams, capacity and therefore prioritization determines the time to delivery. In any time-critical environment, this can become a risk factor that is mitigated by expensive but reliable local solutions. At the same time, relying on central solutions, product teams are forced to accept and consume design that has been defined under conditions and assumptions that are not applicable to their specific domain. This misalignment between central teams and product teams was noted by one product designer, who explained that “central designers don’t have such deep knowledge of customer requirements as the line of business designers.” This inherently holds the risk of unwanted effects and short-term fixes. The main reason for this issue is twofold: lacking or insufficient communication structures between central and de-central teams, and missing procedures for product teams to contribute to central deliveries.

Skills required to specify components differ from the skills of a product designer. While the later requires a deep understanding of individual domain and use cases, the component design must be able to generalize across as many use cases as possible to ensure reusability. These conflicting priorities must be moderated and channeled into a collaborative platform that supports the exchange and negotiation as well as means for flexibility as part of the design system evolution.

4.3 Flexibility

Another way to avoid costly local solutions would be incremental enhancements or extensions of existing solutions. This requires the structural enablement of such extension mechanisms in the design and development processes, which were available only at a comparably high cost. As stated by one of the product designers “each product has unique aspects, and central designers are too far from product specifics. Standards on colors, icons etc. is important - but should not have component level interactions completely standardized - rather a guideline allowing some flexibility.” If a required feature is not available in the design system, product designers should be able to extend it either for local usage or even contributing it back into the design system for common use without creating additional maintenance load onto the teams. Quality and reliability of such extensions must be ensured which requires the skills and mechanisms to sustain this. The main reason the design system did not offer the required flexibility was due to the tight coupling to technology that would support modular extensions only at a comparably high cost. Also, the required skills for creating extension, i.e. system design skills, were not sufficiently available in the product teams.

4.4 Efficiency

The main purpose of design systems and their supporting tool infrastructure is to streamline and accelerate the product design process. This is especially critical due to the permanent lack of design resources, so that designers' time is spent on tasks they can create most value out of, i.e., working with users on understanding their needs and optimizing the design to the tasks at hand. Due to the shortcomings mentioned above, product designers spend a significant amount of their time evaluating technical feasibility, gathering information on available solutions, negotiating with the central teams and eventually creating local design solutions. As mentioned by one of the developers, "sometimes the use case is so specific that you need to develop your own UI element or extend an existing one." The global introduction of moderns, cloud-based design tools that support the distribution of component libraries and themes in a centralized and reliable way has made a big difference compared to the fragmented, often local tool landscape in the past. Still, efficiency gains can only be made when designers can consume components provided as part of the global component set. The integration of the tooling infrastructure and the design system process is therefore essential to overcome the fragmentation of knowledge, componentry, and documentation.

5 Outcome

These four themes helped us to lay out a strategy to improve design system adoption by addressing some of the main inhibitors. We decided to focus on three measures that all would contribute to the improvement along those dimensions.

- **Documentation:** The knowledge gap was mentioned as one of the most important inhibitors. Therefore, we decided to restructure the design system documentation providing a documentation platform that could be used by central teams as well as for enhancements by product design teams. This was backed by additional research.
- **Technology:** The technical infrastructure for user interface elements was not optimized for a modular deployment model supporting contributions and extensions. Moving to a more modular technology (i.e. Web Components) and a thorough decoupling of presentation layer from programming model helped preparing for a contribution model required to become more flexible and responsive to new requirements.
- **Process:** It had become clear during the analysis, that keeping the design system in exclusive ownership of a central team does not scale. To transfer the ownership for the design system into the hands of the entire design community while ensuring quality and reliability would be the most challenging part with the highest impact.

While the change of documentation and technology were more of tactical nature, the transformation of the design system into a community-owned model turned out to be a fundamental culture shift that, while still in process, has already changed the culture and communication within the organization. We decided to employ an adoption of the open-source methodology for organization-internal processes, called inner source.

5.1 Inner Source Design System

Inner source has been established at SAP as a common practice for development projects already [11]. It serves as mechanism to allow contributions by consuming teams into a project owned by another team. As underlined by one of the experts from open-source office, consultancy on inner source is not limited to development teams, and aiding the design system team worked to support it, "Inner source has matured within SAP software development (...) with the intention of contributing across teams." We had the opportunity to work together with internal coaches that were specialized in supporting teams in the adoption of the process. Accounting for the need for flexibility identified in our research we opted for a governance model of shared responsibility where the design system will be owned by multiple teams. In collaboration with all product design teams, we established the following foundational concepts for the design system.

5.1.1 Ownership. The concept of ownership is a central part of the design system to establish responsibility and accountability. Every design team in the organization can become owner for a design system element (e.g., control, pattern, or floorplan). As owner, the team can decide on the scope of an element, can accept or reject the request for additional features, and create a roadmap for the evolution of an element. At the same time, the owner must make sure, the element is documented in the design system portal, specifications are available, and all quality criteria are met (we offer three quality levels, see below). The owner also is responsible for the maintenance of the element jointly with the development team (which can be the central team or a local product engineering team).

5.1.2 Design System Experts (DSE). The design system experts (DSE) are senior design representatives from all product areas. They have multiple responsibilities: they take part in all decisions regarding the inner source model and its implementation; they serve as liaison between inner source and their product design team; they are the gate keepers for contributions by their product teams. This extremely demanding role requires a solid understanding of system design as well as product design requirements.

5.1.3 Elements. The basic entity of the design system are so-called elements. An element is a part of the design system that can be owned by a team. As an initial assumption we decided to not make a difference between simple components, composite components or patterns or even full-page floorplans and navigations. However, we differentiate between common elements that should be used by all teams (e.g. basic components like a button) and other elements that are often more domain-specific, optional and sometimes even experimental. Common elements must adhere to high quality standards and need to be stable and reliable, therefore ownership in most cases remains with the central team. Changes must be aligned with the DSE to identify and avoid side effects.

5.1.4 Foundations. To guarantee the coherence and consistency of the design system, common rules are established as foundations that every element must adhere to. This includes aspects like interaction or value states, accessibility rules, as well as interaction foundations like wrapping and truncation. Foundations are managed centrally

and required alignment across the DSE. Certain foundational aspects are delivered in code as part of the infrastructure to ensure global availability, like for instance design tokens for the design language.

5.1.5 Processes. Design system contributions and requests are managed through a central process implemented with Jira³ which is widely used to manage the design and development process. Requestors can use this to submit requests for enhancements and the requests will be dispatched to the responsible owner or rescope if the requestor was not aware of the actual impact of a request. Owners can use the process to inform about planned changes and where needed request additional perspectives, support, or where needed, approval by the DSE. This process ensures transparency and established reliability for all consumers of the design system. Subsequent development and design activities will then be linked to the original requests to allow tracking along the entire process from discovery to delivery.

5.1.6 Extensibility. The extensibility concept is another important aspect controlling the accumulation of design and development debt. When requesting a feature from a control owned by another team, the ideal solution would be the owner would accept it as an enhancement making it part of the element. The process allows also for various collaboration models between owner and requestor if the final responsibility remains with the owner. If an owner does not want to include a feature, the requestor can request an extension which means that only the additional feature is owned by the requestor while the main part of the element remains with the owner unchanged. This can but does not have to be reflected by the implementation using extensibility concepts like slots. Only as the last solution, teams would be able to fork the element taking full ownership of the detached design. This approach offers a compromise reducing debt but giving the freedom to teams under pressure to deliver without leaving the system.

6 Discussion

The introduction of a community-owned design system making use of the inner source methodology was one of three measures we took to improve the adoption of the design system. While improved documentation and technology were tactical means we could take, the switch to a new working model turned out to be impactful on multiple levels. By tightly integrating the product design teams into the process design, motivational and structural issues could be addressed immediately. Being able to own parts of the design system creates confidence to be able to mitigate delivery pressure and avoiding critical dependencies. Argumentation regarding shortcomings and slow evolution of the design system cannot be kept up as at any time, contributions can be made. This shift also has proven to be successful in changing the perception of the design system and the general acceptance.

However, this also reveals the persisting capacity and skill issues associated to the maintenance and evolution of the design system. Product design teams have their focus on delivering products, where component design is only a means to an end. What should be a motivation investing into component design for a product team?

Taking care of high-quality components that need to fulfil enterprise requirements requires specific skills that are rare in product design teams. Building out the design system expertise inside product design teams and making this also a career path therefore will be one of the next steps we will take. Also, creating communities of interest around certain design system topics (e.g. joining forces on defining best practices in adopting certain technologies like artificial intelligence) will be another instrument to increase the value for product design teams.

One important lesson we have learned, is the fact that the design system experts have become such a critical part in the process that they struggle with integrating this into their day in addition to other tasks they have. This role in most cases cannot be fulfilled by a single person but instead should be covered by a group of designers from the same product area. Establishing this role together with the required training and enablement mentioned above will be another important step to better scale.

Today, not all elements of the design system have defined ownership, so that we had to introduce a concept of temporary ownership that allows teams to realize a required feature and deliver specifications, guidelines, and tools without taking permanent ownership of the element. This might impact consistency of elements over time so that we might need to integrate additional quality gates for such cases.

Handling dependencies between design system elements is still one of the challenges of the approach we need to elaborate further on: we need to ensure the implementation of changes in foundations across all elements (for instance new accessibility requirements), e.g., by centrally generating design tasks for each element; we need to manage dependencies between base components and extensions within the tooling, documentation and implementation, e.g., by providing additional meta data and structural elements.

As result of the initial pilot phase, we have already gone through a major simplification of the overall process. At the same time, we extend it with specific use cases and required quality gates in collaboration with the community.

The transition to a community-based approach had a huge impact on the central design system team. While still being organizationally responsible for the design system, the team had to shift to the role of a community moderator and facilitator in addition to keeping up the design ownership for a large part of the components. The additional efforts that are required to establish and maintain standards and processes, keeping the community alive and enabling the product design teams to become part of the design system evolution are investments that initially reduced the capacity of the central team and that can only pay off in the long term.

Currently, a baseline study is run that will help monitor the impact the approach will have on the efficiency and effectiveness of the design system. This baseline study collects information about the awareness, satisfaction, participation as well as the completeness of the design system. Quantitative data is continuously collected by tracking the request pipelines. Over time, this will help quantify changes in adoption, efficiency, and coverage.

³Atlassian Jira: <https://www.atlassian.com/software/jira>

7 Conclusion

While being still in pilot phase, long-term effectiveness and scalability of the proposed solutions still needs to be determined. However, introduction of the inner source process for the design system has already changed the perception in the design community. A cultural change from merely consuming to actively participating in the design system is gradually taking place within the organization. Changing to a community model does not solve existing resource limitations but creates more flexibility and organizational resilience as teams are starting to understand their role and contribution. This approach still builds on a strong and neutral central team, but as the role shifts away from being the bottleneck to being a partner in achieving common goals, this also improves motivation and sentiments between teams. At SAP we plan to further strengthen the inner source process building out a strong community around design system.

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APPENDICES

A.1 Ethics and Safety Protocol

Protecting the security and privacy of participants was a top priority. To ensure a safe environment for open discussion, several agreements were made: 1) participants and researchers could freely discuss any topic related to the design system in a workspace, 2) participants would not face penalties for their statements. Anonymity was maintained for any unintentional mention. One-on-one virtual interviews were chosen to avoid issues common in group settings, such as participants hesitating to speak due to authority figures. Participants were informed they could withdraw at any time, but none chose to do so. Many expressed enjoyments, reflecting positively on their voices being heard and were pleased to contribute to the research.